Algebra 1
Unit 5
Lesson 10 Practice

Name_Answer Key________
Date___________________
Period_________________

1. Solve the system of equations using elimination.

$$\left(\begin{cases} 2x + 3y = 9 \\ \frac{1}{2}x + \frac{2}{3}y = 4 \end{cases}\right) - 4$$

$$2x + 3y = 9$$
$$-2x - \frac{8}{3}y = -16$$
$$\frac{1}{3}y = -7$$
$$y = -21$$

$$2x + 3(-21) = 9$$
$$2x - 63 = 9$$
$$2x = 72$$
$$x = 36$$

$(36, -21)$

2. Verify the solution to question 1 algebraically using substitution.

$2(36) + 3(-21) = 72 - 63 = 9$ ✓

$\frac{1}{2}(36) + \frac{2}{3}(-21) = 18 - 14 = 4$ ✓

Pam has two part time jobs. At one job, she works as a cashier and makes $\$8$ per hour. At the second job, she works as a tutor and makes $\$12$ per hour. One week she worked 30 hours and made $\$268$. This situation can be represented by the system of equations, $\begin{cases} 8c + 12t = 268 \\ c + t = 30 \end{cases}$, where c is the number of hours she worked as a cashier and t is the number of hours she worked as a tutor.

3. The system could be solved using both substitution or elimination. Which method do you prefer to use?

Elimination

4. Solve the system of equations using your preferred method.

$$-8(c + t = 30) \rightarrow -8c - 8t = -240$$

$$8c + 12t = 268$$
$$+ -8c - 8t = -240$$
$$4t = 28$$
$$t = 7$$

$$c + 7 = 30$$
$$c = 23$$

$(23, 7)$

5. Interpret your solution.

Pam worked 23 hours as a cashier and 7 hours as a tutor.

6. Explain whether negative values for c and t are viable in this context.

No, it does not make sense to work a negative number of hours.

The Johnson and Green families went to a fast-food restaurant for lunch. The Johnson's bought 4 cheeseburgers and 3 medium fries for $\$16.53$. The Greens bought 5 cheeseburgers and 4 medium fries for $\$21.11$. This situation can be represented by the system of equations shown where c represents the cost of a cheeseburger and f represents the cost of medium fries.

$$\left(\begin{cases}4c+3f=16.53\end{cases}\right)-4 \rightarrow -16c-12f=-66.12$$
$$\left(\begin{cases}5c+4f=21.11\end{cases}\right)3 \rightarrow 15c+12f=63.33$$

7. Explain why substitution is not an efficient method to solving the system.

It would take more than one step to isolate a variable and there would be a lot of decimals.

8. Solve the system of equations.

$$\begin{array}{l} -16c-12f=-66.12 \\ \underline{+15c+12f=63.33} \\ -c=-2.79 \\ c=2.79 \end{array}$$

$$\begin{array}{l} 4(2.79)+3f=16.53 \\ 11.16+3f=16.53 \\ 3f=5.37 \\ f=1.79 \end{array}$$

$(2.79, 1.79)$

9. Interpret the solution.

The cost of a cheeseburger is $\$2.79$ and the cost of medium fries is $\$1.79$.

10. Determine the constraints on c and f in this context.

(It does not make sense for cost to be negative)

$0 \leq c \leq 25$
$0 \leq f \leq 25$

(It also does not make sense for the cost of a burger/fries at a fast-food restaurant to be too high)